

Name:- Vaishnavi rajesh bomnale

PRN:-202501050063

Batch :- CH3

Sub :- EDS

Screenshot of practical no 1:

Practical lab assignment:

The image displays two screenshots of the CODETANTRA online IDE interface, showing Python programs and their test results.

Top Screenshot: 1.1.1. Calculate Momentum

Problem Statement: Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

- m is the mass of the object (in kilograms).
- v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases:

Expected output	Actual output
5.0	5.0
10.0	10.0
50.00kgm/s	50.00kgm/s

Code Snippet:

```
1 #Type Content here...
2 mass = float(input())
3 velocity = float(input())
4 momentum = mass * velocity
5 print(f"momentum:.2f)kgm/s")
```

Test Results: 2 out of 2 shown test case(s) passed, 2 out of 2 hidden test case(s) passed.

Bottom Screenshot: 1.1.2. Conditional Calculation Based on the Number of Digits

Problem Statement: Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints: $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases:

Expected output	Actual output
9	9
81	81

Code Snippet:

```
1 #write your code here...
2 import math
3 n = int(input())
4 if 1 <= n <= 9:
5     print(n ** 2)
6 elif 10 <= n <= 99:
7     print(f"{math.sqrt(n):.2f}")
8 elif 100 <= n <= 999:
9     print(f"{n ** (1/3):.2f}")
10 else:
11     print("Invalid")
```

Test Results: 4 out of 4 shown test case(s) passed, 3 out of 3 hidden test case(s) passed.

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY - MM - DD`.
- The second line contains the second date in the format `YYYY - MM - DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

```
1 from datetime import date
2
3 #write your code here...
4 from datetime import datetime
5 date1 = input().strip()
6 date2 = input().strip()
7 d1 = datetime.strptime(date1, "%Y-%m-%d")
8 d2 = datetime.strptime(date2, "%Y-%m-%d")
9 days_between = (d2 - d1).days
10 print(days_between)
```

Average time

0.017 s

Maximum time

0.030 s

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

2014-07-02

2014-11-02

123

Actual output

2014-07-02

2014-11-02

123

Test case 2

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 #write your code here...
2 num = input().strip()
3 n = int(num)
4 rev = 0
5 while n > 0:
6     digit = n % 10
7     rev = rev * 10 + digit
8     n = n // 10
9
10 print(rev)
```

Average time

0.005 s

Maximum time

0.007 s

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1

Expected output

5367

7635

Actual output

5367

7635

Test case 2

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
1 #write your code here...
2 num = int(input())
3 for i in range(1, 11):
4     print(f"{num} x {i} = {num * i}")
5
6
```

Average time

0.019 s

Maximum time

0.056 s

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

Actual output

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

Test case 2

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer `n`, the number of courses.
- The second input will be `n` integers representing the marks of the student in each of the `n` courses, separated by a space.

Output Format:

- If the student passes all courses:
 - Print the aggregate percentage (formatted to two decimal places).
 - Print the grade based on the aggregate percentage.
- If the student fails any course (marks < 40 in any course), print:
 - "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorFa...

```
1 # Number of courses
2 n = int(input().strip())
3
4 # Marks input
5 marks = list(map(int, input().split()))
6
7 # Check for failure
8 if any(mark < 40 for mark in marks):
9     print("Fail")
10
11 else:
12     aggregate = sum(marks) / n
```

Average time

0.008 s

Maximum time

0.011 s

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1

Expected output

55.50 / 78.89

Aggregate: 71.75

Grade: First Division

Actual output

55.50 / 78.89

Aggregate: 71.75

Grade: First Division

Test case 2

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2     # write the code..
3     if n == 1:
4         return 0
5     elif n == 2:
6         return 1
7     else:
8         return fibonacci(n - 1) + fibonacci(n - 2)
9
10
11
```

Average time0.014 sMaximum time0.035 s

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected output0 1 1 2 3Actual output0 1 1 2 3

Test case 2

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 n = int(input().strip())
2 for i in range(1, n + 1):
3     for j in range(i):
4         print("*", end=" ")
5     print()
```

Average time0.006 sMaximum time0.007 s

2 out of 2 shown test case(s) passed4 out of 4 hidden test case(s) passed

Test case 1

Expected output*
* *
* * *
* * * *
* * * * *

Actual output*
* *
* * *
* * * *
* * * * *

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 n = int(input().strip())
2 for i in range(1, n + 1):
3     for j in range(1, i + 1):
4         print(j, end=" ")
5     print()
```

Average time0.006 sMaximum time0.007 s

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected output1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

Actual output1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

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